JavaScript Loops — Exercises

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// Q1: Print numbers from 1 to 10 using a for loop.
console.log("--- Q1 ---");
for (let i = 1; i <= 10; i++) {
  console.log(i);
}

// Q2: Sum of first N numbers - Use a while loop to calculate the sum of the first N natural
numbers.
console.log("--- Q2 ---");
let n2 = 10, sum2 = 0, i2 = 1;
while (i2 <= n2) {
  sum2 += i2;
  i2++;
}
console.log("Sum of first " + n2 + " natural numbers = " + sum2);

// Q3: Multiplication table - Print the multiplication table of a given number using a for
loop.
console.log("--- Q3 ---");
let n3 = 7;
for (let i = 1; i <= 10; i++) {
  console.log(n3 + " x " + i + " = " + n3 * i);
}

// Q4: Factorial calculation - Find the factorial of a given number using a while loop.
console.log("--- Q4 ---");
let n4 = 5, fact = 1, i4 = n4;
while (i4 > 1) {
  fact *= i4;
  i4--;
}
console.log("Factorial of " + n4 + " = " + fact);

// Q5: Reverse counting - Print numbers from 10 down to 1 using a for loop.
console.log("--- Q5 ---");
for (let i = 10; i >= 1; i--) {
  console.log(i);
}

// Q6: Even numbers up to N - Use a do-while loop to print all even numbers up to N.
console.log("--- Q6 ---");
let n6 = 20, i6 = 2;
do {
  console.log(i6);
  i6 += 2;
} while (i6 <= n6);

// Q7: Sum of digits - Calculate the sum of digits of a given number using a while loop.
console.log("--- Q7 ---");
let n7 = 12345, temp7 = n7, digitSum = 0;
while (temp7 > 0) {
  digitSum += temp7 % 10;
```

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  temp7 = Math.floor(temp7 / 10);
}
console.log("Sum of digits of " + n7 + " = " + digitSum);

// Q8: Fibonacci series - Generate the first 10 terms of the Fibonacci series using a for loop.
console.log("--- Q8 ---");
let fib = [0, 1];
for (let i = 2; i < 10; i++) {
  fib.push(fib[i - 1] + fib[i - 2]);
}
console.log(fib.join(", "));

// Q9: Guessing game - Use a do-while loop to keep asking for a number until the user guesses
the correct one.
console.log("--- Q9 ---");
const secret = 7;
const guesses = [3, 9, 5, 7]; // simulated user input
let g = 0, guess;
do {
  guess = guesses[g++];
  if (guess === secret) console.log("Guess " + guess + ": Correct! You win!");
  else console.log("Guess " + guess + ": Wrong, try again");
} while (guess !== secret);

// Q10: Prime number check - Check if a given number is prime using a for loop.
console.log("--- Q10 ---");
let n10 = 29, isPrime = n10 > 1;
for (let i = 2; i * i <= n10; i++) {
  if (n10 % i === 0) {
    isPrime = false;
    break;
  }
}
console.log(n10 + (isPrime ? " is prime" : " is not prime"));
```